Abstract 6004

Real-world clinical actionability of comprehensive genomic profiling in advanced solid tumors: a prospective single-center pan-cancer cohort

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Background

Real-world actionability of comprehensive genomic profiling (CGP) in advanced solid tumors ranges 20-40% across published cohorts, mostly large US/European consortia; Asian single-center first-pass implementation data are scarce. We characterised the biomarker landscape and therapy-match rate at a Taiwanese tertiary cancer center.

Methods

We prospectively enrolled 62 consecutive late-stage solid-tumor patients profiled with a 523-gene panel (TruSight Oncology 500, DRAGEN v2.6) over 8 monthly batches. Tumor mutational burden, microsatellite instability, genomic instability score (HRD proxy), small variants, copy-number alterations and large rearrangements were annotated by OncoKB and CiVIC and tiered by AMP/ASCO/CAP (Li 2017). Therapy-match rates were stratified into OncoKB on-label (Level 1/2/3A) versus off-label (3B/4), with resistance flags (R1/R2) tracked.

Results

Eleven cancer types were represented (endometrial 16, breast 14, colorectal 9, NSCLC 8, ovarian 7, others 8). Biomarker and therapy-match rates appear in the Table. Most recurrent oncogenic driver genes were TP53 (26 patients), KRAS (16), PIK3CA (16), PTEN (11), ARID1A (9), APC (7), ESR1 (6). HRD-positive cases clustered in breast and ovarian; a marquee finding was a BRCA1 large rearrangement (exon 7 LOSS, fold change 0.15) in an HRD-positive breast cancer, supporting on-label PARP inhibition. OncoKB resistance variants (KRAS G12D, NRAS G13D/Q61L) flagged anti-EGFR ineligibility in 6 colorectal cases.

Cohort metric	n / 62 (%)
TMB-High (≥10 mut/Mb)	8 (12.9%)
MSI-High	2 (3.2%)
HRD-positive (GIS ≥42)	6 (9.7%)
≥1 OncoKB on-label match (Level 1/2/3A)	21 (33.9%)
≥1 OncoKB off-label match (Level 3B/4)	38 (61.3%)
≥1 OncoKB resistance flag (R1/R2)	6 (9.7%)

Conclusions

Routine pan-cancer CGP at a Taiwanese tertiary cancer center yielded an on-label therapy match in roughly 1 in 3 advanced solid tumors, consistent with international cohorts. Biomarker-defined subsets (TMB-High, HRD-positive, MSI-High) identified additional immunotherapy- or PARPi-eligible patients. Patient reports openly browseable.

Clinical trial identification

Editorial acknowledgement

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